Welding, Soldering, Brazing

Welding is a metallurgical process joining two pieces of metals by the application of heat with or without the application of pressure and addition of filler metal. The joint formed is a permanent joint.

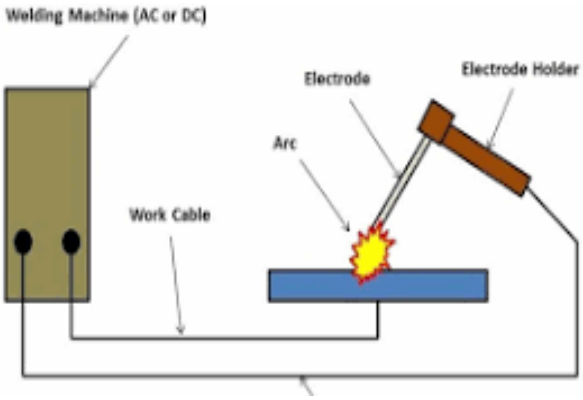
Welding may be classified under two broad headings:

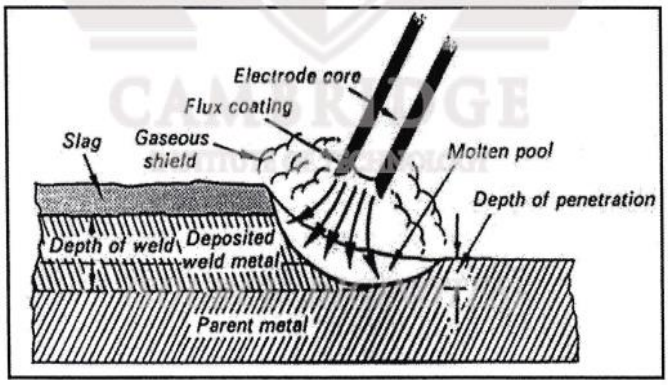
a. Plastic welding process b. Fusion welding process

Plastic welding: In the plastic welding process the pieces of metal to be joined are heated to a plastic state and then forced together by external pressure. This procedure is used in forge welding, resistance welding, spot welding in which pressure is required.

Fusion welding: In the fusion welding, the material at the joint is heated to a molten state and allowed to solidify. This includes gas welding arc welding and Thermit welding.

Arc welding: It is a type of fusion welding. It operates under the principle that when two conductors of an electric circuit are touched together momentarily and then instantaneously separated slightly, assuming that there is sufficient voltage in the circuit to maintain the flow of current, an electric arc is formed. Concentrated heat is produced throughout the length of the arc at a temperature of about 5000 to 6000°C. In arc welding, usually the parts to be welded are wired as one pole of the circuit, and the electrode held by the operator forms the other pole. When the arc is produced the intense heat quickly melts the work piece metal which is directly under the arc, forming a small molten metal of the electrode is carried over by the arc to the molten metal pool of the work piece. The molten metal in the pool is agitated by the action of the arc, thoroughly mixing the base and the filler metal. A solid joint will be formed when the molten metal cools and solidifies. The flux coating over the electrode produces an inert gaseous shield surrounding the arc and protects the molten metal from oxidizing by coming in contact with atmosphere.





Both AC and DC are used for arc welding

For AC arc welding a step-down transformer is used. It receives AC supply between 200 to 440V and transforms it to required low voltage of 80 to 100V. A high current of 100 to 400A. is suitable for arc welding.

In DC arc welding work piece is connected to positive pole of DC generator and the electrode to the negative pole to melt greater mass of metal in the base metal, this setup is called straight polarity. When the heat required is less in the base metal then the polarity is reversed. Due to this option in DC

arc welding it is possible to melt many metals In AC there is no choice of polarity since the current changes every cycle.

Arc welding electrodes:

There are two types of electrodes that are used in arc welding,

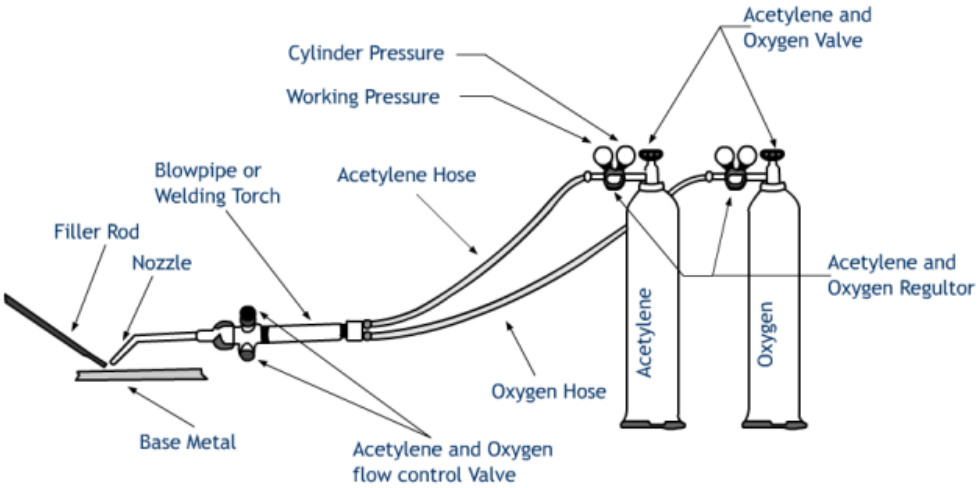
(A) Consumable electrodes (B) Non- consumable electrodes

Consumable electrodes are the electrodes which also melts along with the work piece and fill the joint. Consumable electrodes could be either bare or coated. When bare electrodes are used globules of the molten metal while passing from the electrodes absorb oxygen and nitrogen from atmosphere, which gets trapped in the solidifying weld metal and thereby decreases the strength of the joint. Electrodes are made up of soft steel or alloy steel. The coating consists of chalk, starch, Ferro manganese and binding agents.

Non- consumable electrodes: when these are used, an additional filler material is also required. Advantage in using this electrode is that amount of metal deposited can be controlled which is not possible in other type of electrode.

Gas welding (Oxy-acetylene welding): It is a fusion welding, in which a strong gas flame is used to raise the temperature of the work piece to melt them. Oxy-acetylene gas mixture is most used in gas welding. When right proportions of oxygen and acetylene are mixed in the welding torch flame produced is called as the oxy-acetylene flame. The temperature attained in 3200° C, hence has an ability to melt all commercial metals. The intense heat from the flame melts the base metal, the flame's heat melts the edges of the metals being joined, allowing them to fuse together as they cool.

In many cases, a filler metal rod is used to add material to the weld, strengthening the joint.



Typ parts of oxygen to the parts of the acetylene Depending on the gas ratio following flames are obtained (i) Neutral flame (ii) Oxidizing flame (iii) Reducing flame (carburizing flame).

(i) Neutral flame:

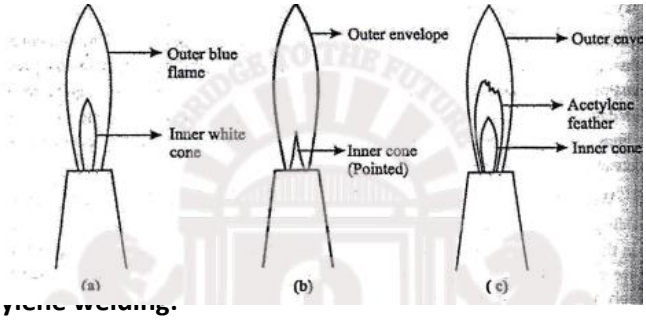
- A neutral flame is obtained by supplying equal volume of oxygen and acetylene
- It consists of a small whitish inner cone surrounded by sharply defined blue flame
- Most of the gas welding is done using the neutral flame

(ii) Oxidizing flame

- This is obtained when there is excess of oxygen, gas ratio
- It appears to be similar to that of neutral flame but the inner white cone flame is shorter than that of neutral flame
- This flame is generally used in metal cutting rather than welding since weld metal gets oxidized

(iii)Reducing flame:

- This is obtained by supplying excess of acetylene in the gas ratio
- It has 3 cones, an inner white cone ,surrounded by an intermediate whitish cone known as " intermediate flame feather" and a bluish envelope flame
- This flame is used for welding alloy steels, cast iron, aluminium



Advantages of oxy acetylene welding:

1. Most versatile process of welding with wide use in various manufacturing process
2. Low cost of the equipment and low cost of maintenance of the equipment
3. Because of separate heat source and filler metal the control can bs exercised on the rate at which the filler metal deposits.
4. The rate of heating and cooling is slow. This help in retaining the structural homogeneity.

Disadvantages: 1. Difficult to attain low-cost target while joining heavy section.

2. Handling and storage of gases not an easy job.
3. It takes long time for the flame to heat up the metal piece than compared to the arc welding.
4. Possible hazards due to explosion of gases.

Soldering

• Soldering is a method of joining similar or dissimilar metals by the application of heat and using a filler metal or alloy called solder, whose liquidus temperature is below 450° C.

• The molten filler metal is made to flow between the two closely placed adjacent surfaces by the capillary action.

• Though soldering obtains a good joint between the two plates, the strength of the joint is limited by the strength of the filler metal used.

Classification of Soldering: Classified into soft soldering and hard soldering.

Soft soldering: Is used extensively in sheet metal work for joining parts that are not exposed to the action of high temperatures and are not subjected to excessive loads and forces or vibrations. Soft soldering is also employed for joining wires and small parts. The solder is mostly composed of lead and tin. In soft soldering, Zinc chloride and ammonium chloride are the most common soldering fluxes used which are quick acting and produce efficient joints. But because of their corrosive nature the joint should thoroughly cleaned of the entire flux residue from the joint. These are to be used only for non-electrical soldering work. Rosin and rosin plus alcohol-based fluxes are least active type and are generally used for electrical soldering work.

Hard soldering: Employs solder which melts at higher temperatures (350° C to 900° C) is stronger than used in soft soldering.

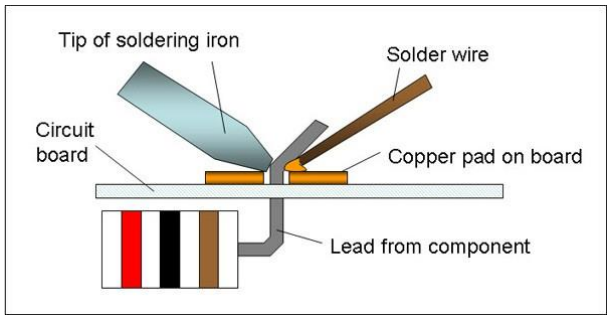
Hard solder is an alloy of copper and zinc to which silver is added sometimes. German silver, used as a hard solder for steel is an alloy of copper, zinc and nickel.

Sequence of operations: The following operations are required to be performed sequentially for making soldered joints.

• **Shaping and fitting of metal parts together**: The two parts to be joined are shaped to fit closely so that the space between them is extremely small and filled completely with solder by capillary action. If a large gap is present, capillary action will not take place, and the joint will not be strong.

• **Cleaning of surfaces**: In order to obtain a sound joint, the surfaces to be soldered are cleaned to remove dirt grease or any other foreign material.

• **Application of flux**: The flux is applied when the parts are ready for joining.



• **Application of heat and solder**: The parts are held in a vice or with special work holding devices so that parts do not move while soldering.

Advantages:

- Simple and economical process.
- Relatively low temperature process, there is no metallurgical damage to base metal.

• The soft soldered joints can easily be dismantled by simple heating.

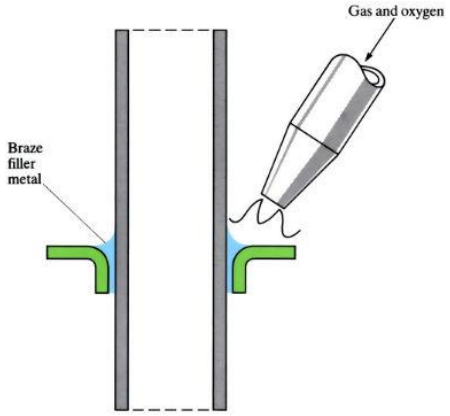
Disadvantage:

• The strength of joint relatively low.

• Flux must be thoroughly cleaned off after soldering, as it is often corrosive.

Brazing

Brazing is a process of making joints where in coalescence is produced by heating to suitable temperatures above 500° C and by using a non-ferrous filler metal having a melting point (up to 900° C) below that of the base metal, the filler metal being distributed between the closely fitted surfaces of the joint by capillary action. Brazing gives a much stronger joint than soldering. Both similar and dissimilar metals can be joined.



The flux along with spe oxides from the surfaces. Borax is the most widely used flux. It will dissolve the oxides of most of the common metals.

The filler material used in brazing is commercially known as spelter. The spelter is usually an alloy of copper, zinc and tin.

Advantages:

• Dissimilar metals, such as stainless steel to cast iron can be joined by brazing. Almost all metals can be joined by brazing except aluminium and magnesium which cannot easilybe joined by brazing.

• Because of the lower temperatures used there is less problems due to heat.

• The joint can be quickly finished without much skill.

• Because of the simplicity of the process it is often an economical joining method with reasonable joint strength.

• The brazed joints are reasonably stronger, depending on the strength of the filler metal used.

Disadvantages:

- Flux material required to prevent the corrosion.
- Large section cannot be joined.
- Fluxes and filler materials may toxic.
- Filler materials are expensive

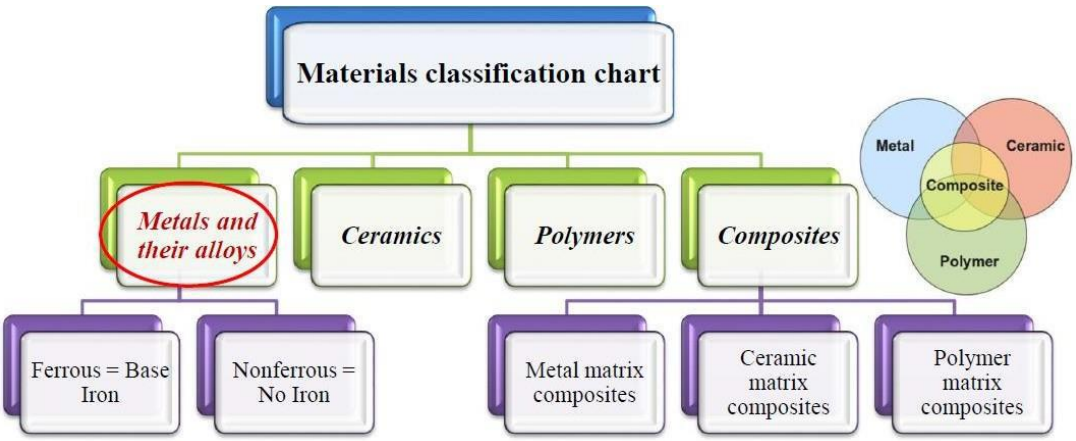
Comparison between Welding, Soldering and Brazing:

Welding	Soldering	Brazing
It is a high temperature process where in the base metals are heated above their melting temperature	Low temperature process, base metals are not melted	The base metals are not melted but broadly heated to a suitable temperature
Filler metals is same as that of base metals	Filler metals is not the same as that of base metals	Filler metals is not the same as that of the base metals
Joint is formed by the solidification of the molten filler metal with the molten base metal	Joints is formed by means of diffusion of the filler metal into the base metal	Joint is formed by means of diffusion of the filler metal into the base metal associated with surface alloying
Strength of weld is much stronger	Strength is low	Lies between welded and soldered joint
Due to high temperature the metal adjacent to the weld portion called heat affected zone is affected to a larger extent	There is no Heat affected zone as the process is carried out at low temperature	Although base metals are heated, the heat affected zone is not too much when compared to welding
Welded joint requires finishing operations	Finishing operation is not required	Little finishing is required
Temperature >3000°C	Temperature <450°C	Temperature >450°C and less than melting point of base metal

Engineering Materials:

Engineering materials are defined as solid substances which are manufactured and used for various engineering applications. There are various materials available and the choice of a particular material for a specific application depends on the set of properties possessed by the materials.

Classification of Engineering Materials



Metals and Their Alloys

- Metal is a material, an element, compound or alloy that is typically hard when in solid state, opaque and shiny
- Metals are generally malleable and ductile

Classification of Metals

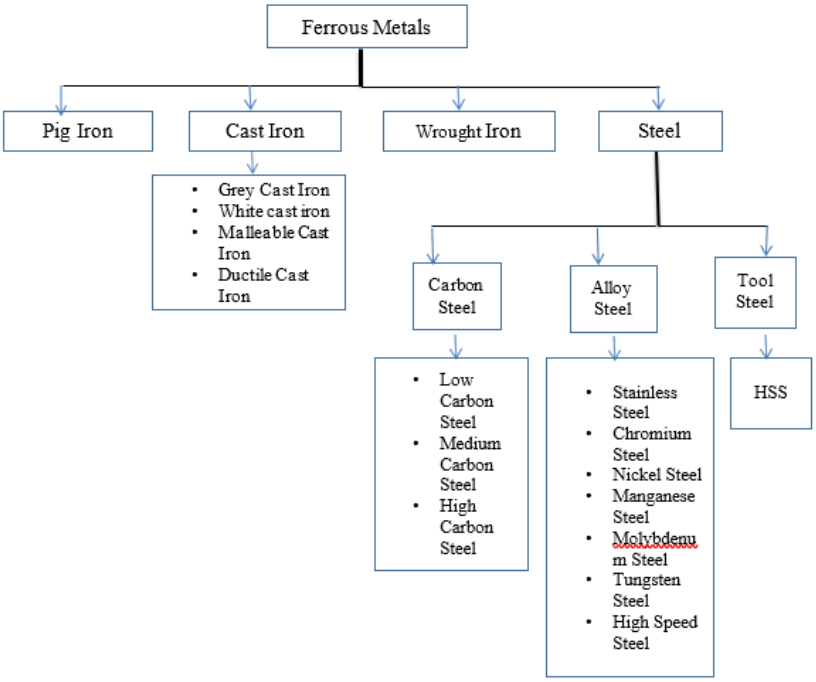
1. **Ferrous Metals:** The term "ferrous" is derived from the Latin word meaning containing "containing iron"

Examples: cast iron, mild steel, medium carbon steel, high carbon steel, stainless steel, and high-speed steel.

2. **Non-ferrous metals:** Non-ferrous metals do not contain iron.

Examples: aluminum, copper, zinc, tin, brass (copper + zinc), and bronze (copper + tin).

Classification of Ferrous Metals



Pig Iron: It is the first stage of iron directly extracted from the ore through blast furnace. It Contain high percentage of carbon and other impurities.

Properties: High hardness and very brittle

Applications: It is used in making wrought iron, cast iron and steel

Wrought iron: Wrought iron is an iron alloy with a very low carbon content

Properties: Low carbon content, Soft, Highly Ductile in nature, High toughness, good machinability, Corrosion resistance, High elasticity

Applications: Due to corrosion resistance, it is used in making steam and water pipes. Decorative applications like fences, gates and railings, balconies, bars, engine bolts, rivets, railway couplings, chain links and crane hooks.

Cast Iron: Is a ferrous material that is produced when pig iron is remelted in a cupola furnace and poured into mould in order to make castings

Properties: Very strong but brittle, Low melting point, Good fluidity, Castability, Good machinability, wear resistance

Applications: Manufacture of machine frames, columns, beds and plates, housing, flywheels, manhole covers, engine block, cylinder head, gearbox case, machine parts etc,

Grey Cast Iron

It is produced when the molten pig iron in the mould is allowed to cool and solidify slowly. Carbon here is mainly in the form of graphite

Properties: Good castabality, High fluidity, easy Machinability, high strength and ductility under compressive loads. It is named as Grey cast iron because of its freshly fractured surface has a grey appearance

Applications: Base structures of machinery, Automotive engine blocks, gears, flywheels, brake discs and drums and machine bases.

White Cast Iron

It is manufactured by casting grey cast iron in a mould and rapidly cooling it. By rapid cooling, there is no free graphite or free carbon and instead there is cementite which gives the cast iron its white colour.

Applications: Used in parts that require high abrasion resistance such as rollers of rolling mills, wear plates, rim of a car wheel, railway wheels brake blocks, hammer mills.

Malleable Cast Iron

It is produced by annealing: a heat treatment of white cast iron. In annealing process white cast iron is heated slowly to a temp about 870oC, and kept for about 25 to 60 hours and then cooled slowly

Properties: It is malleable and can easily machined. Good castability, moderate strength, toughness, corrosion resistant and shock resistant

Applications: Used for automobile parts, motorcycle frames, farm

Steel

Steel is an alloy of iron and carbon with carbon content being less than 2% . In addition, various alloying elements like Sulphur, phosphorous, manganese, nickel, chromium, copper etc., in small proportions.

Steel is classified as

i) Carbon steels

ii) Alloy Steels

iii) Tool Steels

Carbon Steel

Is a kind of steel in which the main alloying constituent is carbon in the range of 0.12-2%.

This tiny amount of carbon produces a material that exhibits great strength, hardness. Other elements present in small proportions are Sulphur, phosphorous, manganese and magnesium

It is classified as

- Low Carbon Steel
- Medium Carbon Steel
- High Carbon Steel

Low Carbon Steel

It is also called as Mild steel and the most common type of carbon steel. It contain 0.03-0.25% of carbon and rest is iron

It is soft, ductile and it has good machinability, good weldability, good toughness, good impact resistance.

Applications: It finds applications in nails, rivets, bolts, nuts, beams, boiler plates, shafts, gears and axles.

Medium Carbon Steel

It has 0.25-0.59% of carbon and rest is iron. It has properties like good toughness, high hardness, high tensile strength, good bending strength, wear resistance and good machinability.

Applications: It is used in transmission shafts, axles, gears, connecting rods, couplings and in spring manufacturing

High Carbon Steel

It has 0.6 – 1.5% of carbon and rest iron.

It has good strength and wear resistance, very ductile, low impact strength.

Applications: It is used in hammers, Chisels, Screws, Punches, Hacksaw blades, Drills, Leaf spring, ball bearings, milling s-cutters

Alloy Steel

An alloy can be defined as a homogeneous mixture of two or more metals. Alloy steel include iron and carbon. These steels are produced by adding other than carbon in calculated amounts in order to provide specific properties.

• Principle alloying elements are chromium, nickel, manganese.

• Alloying elements are added for properties such as ✓ Improved machinability, weldability and castability Improved strength and ductility Improved fatigue and corrosion resistance

• Common types of alloy steels are

✓ Chromium steel

✓ Nickel steel

✓ Manganese Steels

Stainless Steel

• It is an iron-carbon alloy with chromium content of 11.5%-25%depending upon the properties

• Chromium helps to resist corrosion in steels.

• They get the name because they do not corrode or stain easily.

Applications: Manufacturing of kitchen utensils, sinks, food processing equipment, chemical plant and equipment, knife blades, surgical instruments etc.,

Chromium Steel

• In this type of steel chromium is the alloying element

• It improves toughness, resistance to heat, corrosion

Applications: It finds application in balls, rollers, cutting tools

Nickel Steel

In this type of steel Nickel is the alloying element. It contains 3 % nickel, 0.2-0.35% of carbon

Applications:

• It is used in measuring instruments, clock pendulum where coefficient of expansion is very less1

• used in axles, piston rods, components subjected to shocks and fatigue

Manganese Steel

• In this type of steel manganese is the alloying element. It contains 1.5% manganese, 0.4- 0.55% of carbon

Application: Its application includes shafts, gears, axles

Non-Ferrous Metals

These are metals without iron content. These materials are preferred due to

• Good strength to weight ratio

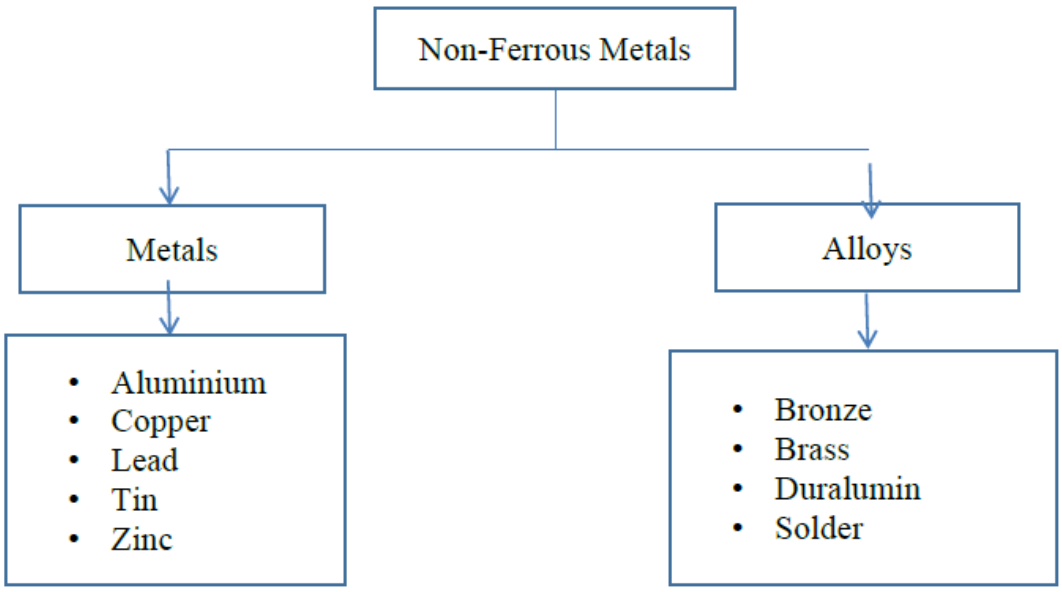
• Good resistance to corrosion

• Light weight

• High electrical and thermal conductivity

• Ease fabrication

Classification



Aluminium

• It is one of the widely used metal in recent years, which has replaced iron and steel because its special properties like light weight, non-corrosive it is very white, soft, ductile material

• It is extracted from Bauxite

• The mechanical properties can be improved by mixing aluminium with silica, copper, zinc

Properties:

• Light weight, high ductility, good electrical and thermal conductivity, good resistance to corrosion

Application:

• Used in making aircraft parts

• Used in automobile for making cylinder block, piston and other engine components

• Used in domestic application for making cooking utensils

• Electrical industry- cables, induction motor, conductors' rotors

Alloys of Aluminium

Duralumin

• It has 92% Aluminium, 3.5-4.5 Copper, 0.4-0.7 % Magnesium, 0.4-7% Manganese

• It is strong as steel but weight only 1/3rd of weight of steel

• Low resistance to corrosion and hence it is coated with pure aluminium to improve corrosion resistance

Application: Used in making connecting rods in automobiles and used in aircraft structures

Magnalumin

• It has 95% Aluminium, 5% Magnesium

• Good damping properties, light weight, high thermal conductivity, nontoxic

• Application:Automobile and aircrafts parts

Y-Alloy

• 93% of aluminium, 4% copper, 2% nickel and 1% magnesium

• Good conductor of heat

• Applications: used to make pistons, cylinder head of IC engine

Copper

• Reddish brown colour, extracted from PYRITE

• It is soft, malleable, ductile, light weight, high thermal conductivity, good corrosion resistance, good electrical and thermal conductivity

Applications:

• Used to make door knobs

• Used in electrical wires, cable, conductors

• Used to make alloy of brass and bronze

• In the form of tubes used in refrigerator, air conditioner and radiator due to high thermal conductivity

Alloys of Copper

Brass

• Is an alloy of Copper and Zinc

• 51-81% of copper, 19-49% of Zinc

• Zinc is added to copper to improve machinability, strength, hardness

• It has good strength, ductility, machinability, good electrical and thermal conductivity, good wear and corrosion resistance

Applications:

• Used in the manufacture of pump parts, marine fittings, valves, condense tubes, fuses, nuts, bolts

• Making ornaments

• Electrical fittings

• Gears, pinions and other moving parts of a clock

• As brazing solder

• Radiator core

• Used in musical instruments such as horn, trumpets

• Bearing material

Bronze

• It is an alloy of Copper and Tin

• Copper 88%, Tin 11% and other alloying elements like phosphorous, lead

• Good strength compare to brass

• Good hardness, easy machinability, good fatigue strength, good corrosion resistance

• Popular type of bronze include Gunmetal and Bell Metal

Gun Metal

• 88% of Copper, 10% Tin, 2% Zinc

• Applications in marine fittings, pumps, valves, bearings, bushes

Bell Metal

• 75-80% Copper and remaining Tin

• Good ringing and low damping quality and it is used to make bells

Lead

• Is a soft and malleable metal obtained from its ore GALENA

• It has poor tensile strength, high coefficient of thermal expansion, it is toxic in nature

• It is soft, ductile and mellable

Applications

• Solder since it has low melting point

• Grid/plate in lead acid batteries

• Pain industry as a colouring pigments

• Water pipes due to its corrosion resistance

Alloys of Lead

• **Solder:** Lead and tin used for joining less fusible metals

Lead foil: is made sandwich of lead between two sheets of tin,. Lead foils are used in radiation shielding

• **Lead-antimony alloy:** are used in storage batteries grid

Zinc

• Is heavy bluish white metal which can be extracted from Zinc Sulphide or Zinc blend

• It is fair conductor f electricity

• It has low melting point and boiling point

• It is resistance to corrosion and good castability

Application

• Galvanizing: To protect iron and steel form corrosion, they are immersed in a bath of liquid zinc or electroplated with zinc, this is called galvanizing

• Galvanised steels is used in produce automobile parts

Alloy of Zinc

• Cadmium zinc alloy

• Magnesium zinc alloy

• Copper zinc alloy

• Lead zinc alloy

• Iron zinc alloy

Tin

• It is a silvery white metal obtained from Tin stone

• It is soft, malleable and ductile, low melting point

Applications

• Tin is coated on steel containers for storing food/ water

• Used a n alloying element in soft solder

Silica

• **Silicon** is a chemical element with the symbol Si and atomic number 14.

• It is a hard, brittle crystalline solid with a blue-grey metallic luster and is a tetravalent metalloid and semiconductor.

• It is relatively unreactive. Its oxides form a family of anions known as silicates.

• Its melting and boiling points of 1414 °C and 3265 °C, respectively, are the second highest among all the metalloids and nonmetals, being surpassed only by boron.

• Silicon is the eighth most common element in the universe by mass, but very rarely occurs as the pure element in the Earth's crust.

• More than 90% of the Earth's crust is composed of silicate minerals, making silicon the second most abundant element in the Earth's crust (about 28% by mass), after oxygen.

Types of Silica

1. Amorphous Silica

a) Natural

b) By-products- Fused Silica, Silica Fume

c) Synthetic Amorphous Silica- Silica gel, colloidal silica, fumed silica

2. Crystalline Silica

a) Cristobalite b) Quartz c) Tridymite

Applications of Silica

• Most silicon is used commercially without being separated, often with very little processing of the natural minerals.

• Silica is used in industrial construction with clays, silica sand, and stone.

• Silicates are used in Portland cement for mortar and mixed with silica sand and gravel to make concrete for walkways, foundations, and roads.

• Used in whiteware ceramics such as porcelain, and in traditional silicate-based soda-lime glass and many other specialty glasses.

• Silicon compounds such as silicon carbide are used as abrasives and components of high-strength ceramics.

• Silicon is the basis of the widely used synthetic polymers called silicones.

Glass

• **Glass** is a non-crystalline, often transparent, amorphous solid that has widespread practical, technological, and decorative use.

• Glass is most often formed by rapid cooling (quenching) of the molten form; some glasses such as volcanic glass are naturally occurring.

• The most familiar, and historically the oldest, types of manufactured glass are "silicate glasses" based on the chemical compound silica (silicon dioxide, or quartz), the primary constituent of sand. Soda–lime glass, containing around 70% silica, accounts for around 90% of manufactured glass.

• The term *glass*, in popular usage, is often used to refer only to this type of material, although silica-free glasses often have desirable properties for applications in modern communications technology.

• Despite being brittle, buried silicate glass will survive for very long periods if not disturbed, and many examples of glass fragments exist from early glassmaking cultures.

The following are the types of glasses,

(1) Soda-lime glass or commercial glass

(2) Potash-lime glass

(3) Potash-lead glass

(4) Common glass

(5) Borosilicate glass.

Application of Glass

• Window panes, tableware, and optics.

• In insulation, sound deadening, as fillers in plastics, and as reinforcement in plastic laminates and structural shapes.

• Glasses can be used to transmit or shield radiation.

• Photosensitive glasses for fluidic devices are used in machine controls.

• Glasses are widely used in the food industry because their chemical resistance and many foods are processed in glass-lined tanks.

• Glass has been used in building construction since ancient time.

Graphite

• Graphite is a crystalline form of the element carbon. It consists of stacked layers of graphene.

• Graphite occurs naturally and is the most stable form of carbon under standard conditions.

• Synthetic and natural graphite are consumed on large scale for uses in pencils, lubricants, and electrodes.

• Under high pressures and temperatures, it converts to diamond. It is a weak conductor of heat and electricity.

Classifications of Graphite

Graphite can be divided into two main types—natural and synthetic.

Natural Graphite

• Natural graphite is a mineral composed of graphitic carbon. It varies considerably in crystallinity.

• Most of the commercial (natural) graphites are mined, and typically contain other minerals. After graphite is mined, it usually requires a considerable amount of mineral processing like froth flotation to concentrate the graphite.

• Natural graphite is an excellent conductor of heat and electricity, stable over a broad range of temperatures, and a highly refractory material with a high melting point of 3650 °C.

• There are three types of natural graphite:

a) High crystalline

b) Amorphous

c) Flake

Crystalline Graphite

• It is said that crystalline vein graphite came from crude oil deposits that have transformed into graphite through time, temperature, and pressure.

• Vein graphite fissures typically measure between 1 cm and 1 m in thickness and usually have a purity of more than 90%.

Amorphous Graphite

• Amorphous graphite is the least graphitic among the natural graphites.

• Amorphous graphite can be found as minute particles in beds of mesomorphic rocks such as coal, slate, or shale deposits.

• The graphite content varies from 25% to 85% according to the geological environment.

• Conventional mining techniques are used to extract amorphous graphite, which occurs mainly in Mexico, North Korea, South Korea, and Austria.

Synthetic Graphite

• Synthetic graphite can be produced from coke and pitch.

• Although this graphite is not as crystalline as natural graphite, it is likely to have higher purity.

• There are basically two types of synthetic graphite.

• One is electrographite, pure carbon produced from coal tar pitch and calcined petroleum coke in an electric furnace.

• The second is synthetic graphite, produced by heating calcined petroleum pitch to 2800 °C.

• Synthetic graphite has higher electrical resistance and porosity, and lower density. Its enhanced porosity makes it unsuitable for refractory applications.

• Synthetic graphite contains mainly graphitic carbon that has been attained by graphitization, heat treatment of non-graphitic carbon, or chemical vapor deposition from hydrocarbons at temperatures over 2100 K.

Applications of Graphite

Refractory Materials

Chemical Industry

Nuclear Industry

Electrical Applications

Other Applications

Amorphous graphite is used in:

• Metallurgy

• Coatings

• Lubricants

• Refractories

• Paint production

• Pencil production

Areas where crystalline graphite can be used include:

• Lubricants

• Powder metallurgy

• Grinding wheels

• Batteries

Diamond

• Diamond offers many advantages over other wide-bandgap materials and thus is a very important material in engineering applications.

• It can be used in high-speed electronics and response systems as well as high-power laser windows, protective coatings, electrochemical sensors, and more.

• Diamond is an allotrope of carbon in which the carbon atoms are arranged in a diamond cubic crystal lattice.

• Of all naturally occurring materials, diamond boasts the highest thermal conductivity and hardness.

• These properties of diamond make it a very useful component of industrial cutting and polishing tools.

• Diamond is the allotrope of carbon in which the carbon atoms are arranged in the specific type of cubic lattice called diamond cubic.

• It is a crystal that is transparent to opaque and which is generally isotropic (no or very weak birefringence).

• Diamond is the hardest naturally occurring material known. Due to important structural brittleness, bulk diamond's toughness is only fair to good.

Uses of Diamond

Diamond has a wide range of applications in several industries

Uses of Diamond in Jewellery

They are used in jewellery (such as earrings, nose rings, engagement rings, pendants, etc.) due to their shiny lustre and their durability. They are used in making jewellery because of their durability and lustre property

Uses of Diamond in Industry

Diamonds are used for industrial purpose. Its property of hardness is useful to drill, grind or cut materials. Hence, some blades used for cutting and drills in the industry used diamonds. They are present on the edges and tips in small sizes.

Applications of Diamond

The applications of diamonds have grown enormously, exploiting the unique combination of properties and the increased availability of the material as synthesis methods developed.

The following applications are covered in more detail below:

Cutting Tools and Wear Components

• The properties exploited by these applications are hardness, strength, low thermal expansion coefficient, low friction coefficient and chemical resistivity.

• Some of the products in this area include oil drilling bits, rock drill cutters, wire drawing dies, extrusion dies, cutting tool inserts, optical grinding tools, coatings for computer hard discs and coatings for ball bearings.

Thermal Management in Substrates, Heat Spreaders and Heat Sinks

Diamond provides the unique combination of high thermal conductivity with electrical insulation.

Semiconductor Devices

The electronic structure of diamond has a wide band gap giving it the potential for use as a semiconductor

Optical Components

Diamond is starting to be used in optical components, particularly as a protective coating for infrared optics in harsh environments.

High-Performance Applications of Diamond

The excellent properties of diamond have the potential to improve performance in many applications. The main barriers are cost and scale of production. For example, metal wires and fibres can be CVD coated with diamond increasing the modulus close to that of diamond and opening the possibility of stronger and stiffer composites, ceramic armour and projectiles

Shape-memory alloy

Shape-memory alloy (SMA) is an alloy that can be deformed when cold but returns to its pre-deformed ("remembered") shape when heated.

• It may also be called memory metal, memory alloy, smart metal, smart alloy, or muscle wire.

• Parts made of shape-memory alloys can be lightweight, solid-state alternatives to conventional actuators such as hydraulic, pneumatic, and motor-based systems.

• They can also be used to make hermetic joints in metal tubing.

Types of Shape-memory alloy

• The two most prevalent shape-memory alloys are copper-aluminium-nickel and nickel titanium (NiTi), but SMAs can also be created by alloying zinc, copper, gold and iron.

• Although iron-based and copper-based SMAs, such as Fe-Mn-Si, Cu-Zn-Al and Cu-Al-Ni, are commercially available and cheaper than NiTi, NiTi-based SMAs are preferable for most applications due to their stability and practicability as well as their superior thermo-mechanic performance.

Applications of Shape-memory alloy:

• Recent research of SMA has been applied in the field of Aerospace, Automotive, Biomedical, and Robotics.

• This memory effect is due to the presence of austenite and martensite crystalline structures. These alloys are bio-compatible, lightweight, and also possess a high force-to-weight ratio.